

Advanced Web Design 21I.WD2.CDE.3

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Lesson: Working with HTML5 page layout and APIs

Tutorial: Creating Navigation Bars

Introduction

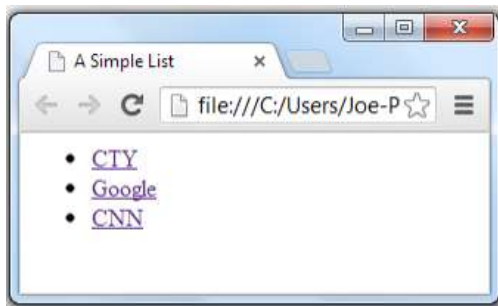
In this tutorial you will learn how to take a simple unordered list element and, using CSS, turn it into both vertical and horizontal navigation bars that can be used to link to multiple pages in a website or to link to other websites.

Making a Simple Vertical Navigation Bar

We'll start by developing a simple vertical navigation bar. Consider the following HTML code:

```
<ul>  
  <li><a href="http://www.cty.jhu.edu">CTY</a>  
  <li><a href="http://www.google.com">Google</a>  
  <li><a href="http://www.cnn.com">CNN</a>  
</ul>
```

The code creates a simple unordered list of three items, with the items being links to three different web pages. When a web page containing this code is displayed in a browser, it looks like this:



Now what we're going to do is to take this very simple code and turn it into a neat little navigation bar by styling the elements using CSS. And...the really cool thing is...we're not going to touch this HTML code at all.

We're going to style the list items to give them the appearance of clickable buttons. We'll develop our solution in a series of steps so that you can understand exactly what is happening.

1. The first thing we'll do is to get rid of the bullets on the list items. This can be accomplished by using the **list-style-type** property and setting its value to **none**, as illustrated below:

```
list-style-type: none;
```

2. Then, we'll make each list item the same width, because we want the bullets all to be the same size. We can accomplish this using the **width** property, and we'll give the buttons a width of **7em**, as illustrated below:

```
width: 7em;
```

3. We'll want the link text to be centered in each button, so we'll set the **text-align** property value to **center**, as illustrated below:

```
text-align: center;
```

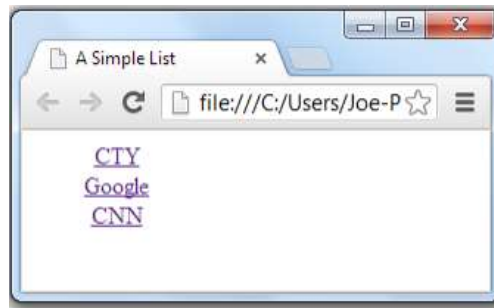
4. And there's one more thing we'll need to do. Recall that list items are automatically indented from the left margin. Actually, they're indented about 2.5em from the left. We don't need or want any indentation since we're not using any bullets, so we'll override the default indentation by using the **margin-left** property and giving it a value of **-2.5em**, like this:

```
margin-left: -2.5em;
```

5. Now, let's define a **div** element, called **menu**, that will be applied to list items and which contains all of this CSS code, and let it do all the formatting work. We'll then simply apply the div to the unordered list.

```
#menu li {  
    list-style-type: none;      /* Get rid of bullets */  
    width: 7em;                /* Make each element same width */  
    text-align: center;        /* Center text in each button */  
    margin-left: -2.5em;       /* Override default indentation */  
}
```

Now, when our page is displayed in a browser, it will look like this:

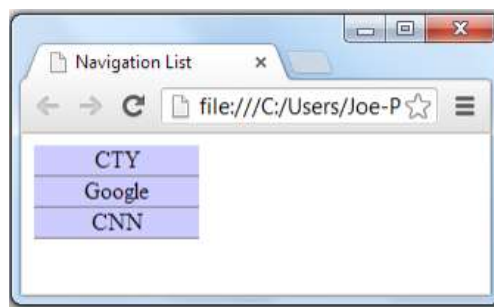


The links don't look like buttons yet, so let's go ahead and take care of that. To make them look more like buttons we're going to style the anchor element, as illustrated below:

```
#menu a {  
    text-decoration: none;      /* Get rid of underscores */  
    color: black;               /* Set text color to black */  
    background-color: #CCCCFF;  /* Make buttons light blue */  
    border-bottom: 1px gray solid; /* Add a bottom border */  
    display: block;             /* Make buttons clickable */  
}
```

In the above code, the first thing we did was to get rid of the default link decoration, which is an underscore. Then, we set the text color to black, and set the button color to a shade of light blue. To get a bit of separation between buttons we've added a bottom border. And the last thing we did was to use a really neat trick to make the buttons clickable...we set the **display** property to have a value of **block**. You've already seen the **display** property used like this. It normally makes an element act like a block level element. However, when it is used like this with an anchor element, it *makes the entire button clickable*...not just the text. How cool is that?

If we add the above code to our web page, it will look like this when displayed in a browser:



Now, the links look more like buttons. There's one more thing we'll do to make the list items appear more like real button controls. We'll style the link state so that when a user hovers a mouse over an item it will appear as if the button is depressed, by adding a thin gray border inset, as follows:

```
#menu a:hover {
    border: 1px gray inset; /* Make button look depressed */
}
```

Exercise

Now, I'd like you to do a simple exercise. I'd like you to implement the CSS and HTML code for the vertical navigation bar and verify that it works correctly.

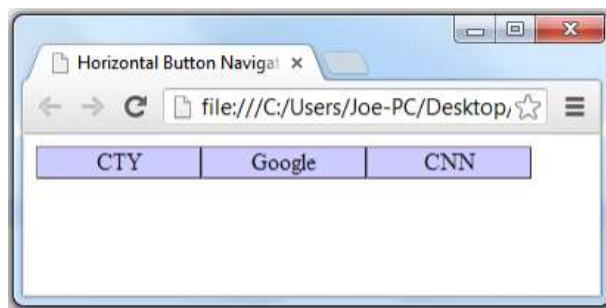
Making a Simple Horizontal Navigation Bar

Now, we're going to change the navigation bar in the last example from a vertical navigation bar to a horizontal navigation bar...and it going to be incredibly simple to do. All we'll need to do is to modify the CSS code slightly. The HTML code remains unchanged. Here's the required CSS code. The changes are displayed in red:

```
#menu ul { margin-left: -2.5em; } /* Move the entire list left */
#menu li {
    list-style-type: none;
    width: 7em;
    text-align: center;
    float: left; /* Float list elements left */
}
#menu a {
    text-decoration: none;
    color: black;
    display: block;
    border: 1px gray outset; /* Put borders all around */
    background-color: #CCCCFF;
}
#menu a:hover { border: 1px gray inset; }
```

In the code above, the following changes were made. First, the entire list is moved to the left to override the default indentation...so we just took the **margin-left** property from the list item styling and moved it into its own style that will be applied to the entire unordered list. Second, using the **float** property, we floated each list item to the left. That will make the items line up left to right. And third, we added an **outset** border all around each button, not just at the bottom, because we'll need to see the separation between the horizontally displayed buttons. That's all there is to it.

And...here's the final result when displayed in a browser:



Exercise

Now, I'd like you to another simple exercise. I'd like you to implement the CSS and HTML code for the horizontal navigation bar and verify that it works correctly.

Summary

This tutorial demonstrated how easy it is to style simple lists using CSS to turn them into vertical and horizontal navigation bars, and demonstrated some of the power of CSS. We were able to get completely different looks without even touching the HTML code.

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